# CBE 150B F2026
# Problem Set 4

Assigned: September 16, 2026
Due: September 23, 2026

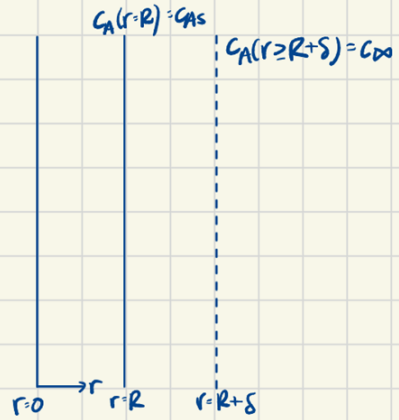
## Problem 1

A biotech company is developing a new pill formulation using a small-molecule drug. The cylindrical pill is designed to dissolve very slowly, allowing for extended release of the desired medication. The pill is a long cylinder of length $L$ and radius $R$. The pill is submerged in a water bath at constant $T$, $P$, and a boundary layer of thickness $\delta$ is established as the drug diffuses radially outward.

The diffusivity of oligosaccharide (pill) in the water is $D_{AB}$. Assume that the pill is impermeable to water, and the surface concentration of dissolved drug is given by the saturation concentration at $T$, $P$: $c_{As}$. Convective mass transport just outside of the boundary layer is infinitely fast and that the bulk water bath has a trace oligosaccharide concentration of $c_\infty$. Assume that the total molar concentration is constant.

**a)** Solve for the steady state concentration profile and flux of cellulose in the boundary layer.

a)




$$\nabla \cdot \underline{N}_A + \frac{\partial C_A}{\partial t} - R_A = 0$$

(steady state; no reaction)

↓ cylindrical

$$\frac{1}{r}\frac{\partial}{\partial r}(r N_{Ar}) + \frac{1}{r}\frac{\partial N_{A\theta}}{\partial \theta} + \frac{\partial N_{Az}}{\partial z} = 0$$

(axial symm; long cylinder)

same assumptions for $H_2O$:

$$\frac{d}{dr}(r N_{Ar}) = 0 \rightarrow C_1 = r N_{Ar} \quad \neq$$

$$\frac{d}{dr}(r N_{Br}) = 0 \rightarrow B_1 = r N_{Br}$$

BCs:

* A: i) $C_A(r=R) = C_{AS}$, ii) $C_A(r=R+\delta) = C_{\infty}$
* B: $N_{Br}(r=R) = 0 \rightarrow$ @ $r=R$, $B_1 = R(0) \therefore B_1 = 0 \rightarrow 0 = r N_{Br} \therefore N_{Br} = 0$

$$\underline{N}_A = -\mathcal{D}_{AB}\nabla C_A + x_A(\underline{N}_A + \underline{N}_B) \xrightarrow{1D} N_{Ar} = -\mathcal{D}_{AB}\frac{dC_A}{dr} + x_A(N_{Ar} + N_{Br}) \rightarrow N_{Ar} = \frac{-\mathcal{D}_{AB}}{1-x_A}\frac{dC_A}{dr}$$

$$\neq \rightarrow C_1 = r\left(\frac{-\mathcal{D}_{AB} C_{tot}}{1-x_A}\frac{dx_A}{dr}\right) \qquad K_1 = \frac{1}{r}dr = \frac{1}{1-x_A}dx_A \rightarrow K_1 \ln(r) = -\ln(1-x_A) + K_2''$$

$$i) \rightarrow K_1 \ln(R) = -\ln\left(1-\frac{C_{AS}}{C_{tot}}\right) + K_2 \; ; \quad ii) \rightarrow K_1 \ln(R+\delta) = -\ln\left(1-\frac{C_{\infty}}{C_{tot}}\right) + K_2$$

$$K_1(\ln(R) - \ln(R+\delta)) = -\ln\left(1-\frac{C_{AS}}{C_{tot}}\right) + \ln\left(1-\frac{C_{\infty}}{C_{tot}}\right) = \ln\left(\frac{1-C_{\infty}/C_{tot}}{1-C_{AS}/C_{tot}}\right) = \ln\left(\frac{C_{tot}-C_{\infty}}{C_{tot}-C_{AS}}\right) = K_1 \ln\left(\frac{R}{R+\delta}\right)$$

$$K_1 = \ln\left(\frac{C_{tot}-C_{\infty}}{C_{tot}-C_{AS}}\right)\left[\ln\left(\frac{R}{R+\delta}\right)\right]^{-1}; \quad K_2 = \ln(R)\ln\left(\frac{C_{tot}-C_{\infty}}{C_{tot}-C_{AS}}\right)\left[\ln\left(\frac{R}{R+\delta}\right)\right]^{-1} + \ln\left(1-\frac{C_{AS}}{C_{tot}}\right)$$

$$\ln\left(1-\frac{C_A}{C_{tot}}\right) = -K_1\ln(r) + K_1\ln(R) + \ln\left(1-\frac{C_{AS}}{C_{tot}}\right) = K_1\left(\ln\left(\frac{R}{r}\right)\right) + \ln\left(1-\frac{C_{AS}}{C_{tot}}\right)$$

$$\boxed{\ln\left(1-\frac{C_A}{C_{tot}}\right) = \ln\left(\frac{C_{tot}-C_{\infty}}{C_{tot}-C_{AS}}\right)\left[\ln\left(\frac{R}{R+\delta}\right)\right]^{-1}\ln\left(\frac{R}{r}\right) + \ln\left(1-\frac{C_{AS}}{C_{tot}}\right)}$$

**b)** Is the flux of cellulose out of the pill constant with respect to r?

b) No ($rN_{Ar}$ is constant)

**c)** If $c_{As}/c_{total} \approx 1$:

**i)** Is there diffusive flux of water near the pill/water interface? In which direction?

c) i) Yes, water diffuses towards pill/water interface (-r direction).

**ii)** Is there convective flux of water near the pill/water interface? In which direction?

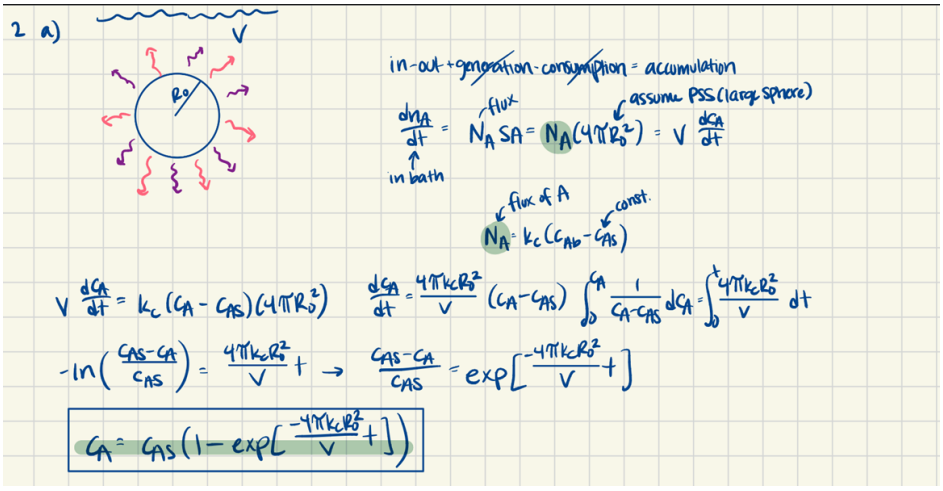
ii) Yes, water moves away from the pill/water interface (+r direction). This maintains $N_B = 0$.

## Problem 2 (XX pts)

A large bath bomb is slowly dissolving in a Jacuzzi bathtub. Assume the bath bomb (A) is surrounded by water on all sides and the jets are on high (dissolution dominated by convection and tub is well-mixed).

You would like to know how quickly the concentration of A increases in the tub. The manufacturer has provided a mass transfer coefficient $k_c$, and the initial radius of the bath bomb is $R_0$. The surface concentration is $c_{As}$, and the bath volume is V.

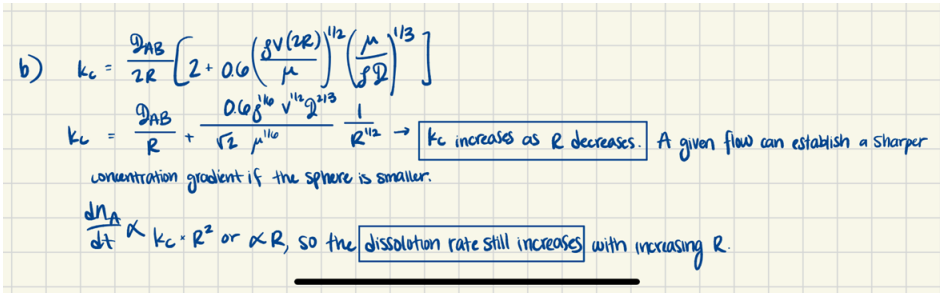
**a)** Perform a mole balance on the bath bomb to derive an expression for the bulk bath bomb concentration in the tub, $c_A(t)$.




For convection around a sphere,

$$Sh = \frac{2k_c R}{D_{AB}} = 2 + 0.6Re^{1/2}Sc^{1/3}$$

**b)** If the bath bomb were small/dissolving quickly, would $k_c$ increase or decrease? What are the implications for the rate of dissolution of the bath bomb?



## Problem 3 (25 pts)

Diamond-based semiconductors are being explored as an alternative to silicon for next-generation microelectronics. Films are deposited under steady state conditions via Chemical Vapor Deposition (CVD) at constant temperature and pressure.

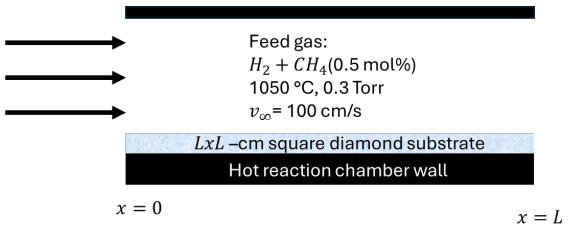
To produce synthetic diamond layers, a hydrogen carrier gas containing dilute methane (0.5 mol%) is flowed via laminar flow over a flat, $L$=10-cm square diamond substrate inside a reaction chamber.

The methane reacts with the substrate, thickening the diamond layer as follows:

$$CH_4(g) \rightarrow C(diamond, s) + 2H_2(g)$$

The reaction is assumed to be rapid such that the deposition rate is controlled by methane mass transport. The reaction chamber is maintained at 1050 $^\circ C$ and a constant pressure of 0.3 Torr. The methane-containing gas stream is flowed in excess such that the bulk gas methane concentration is constant throughout the chamber at a velocity of 100 cm/s parallel to the substrate surface.

The diffusivity (low pressure, high T) at the reaction conditions $\approx$ $35,100$ cm$^2$/s and the viscosity of hydrogen gas $\approx$ $4.40 \cdot 10^{-5}$ Pa $\cdot$ s.




**a)** Where (x-position) along the wafer surface do you expect the film to be the thickest?

a) Conc. boundary layer de e ρ as x ↑, thus the conc. adjacent to the substrate is highest at $\boxed{x=0}$, meaning the deposition rate is highest here.

**b)** Calculate the Schmidt and Sherwood numbers for this situation.

b) $Sc = \frac{\nu}{D_{AB}}$; $Sh = \frac{k_c L}{D_{AB}} = 0.664\, Re^{1/2} Sc^{1/3}$; $Re = \frac{\rho v d}{\mu}$

Assume ideal gas → $PV = nRT \rightarrow \frac{n}{V} = \frac{P}{RT} = \frac{(0.3)\,torr\,(\frac{101325\,Pa}{760\,torr})}{(8.314)\,m^3\,Pa\,(1323.15)\,K\,mol^{-1}\,K^{-1}} = 0.00360359\ \frac{mol}{m^3}$

$\rho = \frac{n \times MW}{V} = \frac{(0.00360359)\,mol\,(2.01)\,g\,kg}{m^3\,(mol)\,(10^3)\,g} = 7.30812E-6\ \frac{kg}{m^3}$

diluto in $CH_4$, so $MW \approx MW_{H_2}$

$Re = \frac{(7.308E-6)\,kg\,(100)\,cm\,(1)\,m\,(10)\,cm\,(1)\,m\,(1)\,Pa}{m^3\ s\ (100)\,cm\ (100)\,cm\ (4.4E-5)\,Pa\,s\,(kg\,m^{-1}\,s^{-2})}$

$= 0.016609$ (Laminar)

$Sc = \frac{(4.4E-5)\,Pa\,s\,m^3\,(1)\,kg\,m^{-1}\,s^{-2}\ \ s\ \ (100^2)\,cm^2}{(7.308E-6)\,kg\,(1)\,Pa\ \ (35100)\,cm^2\,(1^2)\,m^2} = 1.7153 \rightarrow \boxed{1.7 = Sc}$

→ $Sh = 0.664\,(0.016609)^{1/2}\,(1.7153)^{1/3} = 0.102437 \rightarrow \boxed{Sh = 0.10}$

**c)** What do your values of Sc and Sh imply about the magnitude of diffusive mass transport in this system?

c) $Sc > 1$ means that momentum "diffuses" (transfers) faster than $CH_4$ diffuses.
$Sh < 1$ means that $k_c$ (mass transfer coeff.) small relative diffusion length, $D_{AB}/L$. Thus, $CH_4$ must "diffuse far" / mass transport is diffusion-limited.

**d)** Using the local Sherwood number (at a given x) as a function of the local Reynolds and Schmidt numbers (at a given x), evaluate the mass transfer coefficient halfway ($x = \frac{1}{2}L$) and at the end of the substrate ($x = L$).

3. d) $Sh_x = 0.2155\, Re_x^{1/2} Sc^{1/3}$; $k_c = Sh_x D_{AB} x^{-1}$

$Sh_x(x = \frac{1}{2}L) = 0.2155 \left(\frac{(7.308E-6)\,kg\,(100)\,cm\,(5)\,cm\,(1)^2\,m^2}{m^2\ \ s\ \ (100)^2\,cm^2\,(4.4E-5)\,s\,(kg\,m^{-1}\,s^{-2})}\right)^{1/2} (1.7153)^{1/3}$

$Sh_x(x = \frac{1}{2}L) = 0.0235 \rightarrow k_c = (0.0235)(35100)\,\frac{cm^2}{s}\,(5)^{-1}\,cm^{-1} = \boxed{165\ cm/s = k_c(x = \frac{1}{2}L)}$

$Sh_x(x = L) = 0.2155\,(0.6166)^{1/2}\,(1.7153)^{1/3}$

$Sh_x(x = L) = 0.0332 \rightarrow k_c(x = L) = (0.0332)(35100)\,\frac{cm^2}{s}\,(10)^{-1}\,cm^{-1} = \boxed{116\ cm/s = k_c(x = L)}$

**e)** At which point on the plate is the mass transfer coefficient higher and why?

e) $k_c$ higher at $\frac{L}{2}$ because the boundary layer is thinner there. The concentration of methane is higher at the edge of the boundary layer, thus more $CH_4$ is "convected" away.

**f)** Evaluate the average mass transport coefficient ($\bar{k}_c$) for this system from 0 to $L$ (hint: integration).

f) $\bar{k}_c = \frac{\int_0^L k_c\,dx}{\int_0^L dx} = \frac{1}{L}\left[\int_0^L Sh_x \mathcal{D}_{AB} x^{-1}\,dx\right] = \frac{0.2155\mathcal{D}_{AB}Sc^{1/3}}{L}\int_0^L Re_x^{1/2}\cdot\frac{1}{x}dx$

$= \frac{0.2155\mathcal{D}_{AB}Sc^{1/3}}{L}\int_0^L \left(\frac{\rho V x}{\mu}\right)^{1/2}\cdot\frac{1}{x}dx = \frac{0.2155\mathcal{D}_{AB}Sc^{1/3}(\rho V)^{1/2}}{\mu^{1/2}L}\int_0^L x^{-1/2}dx = \frac{0.2155\mathcal{D}_{AB}Sc^{1/3}(\rho V)^{1/2}}{\mu^{1/2}L}\left(2x^{1/2}\right)\Big|_0^L$

$= \frac{0.2155\mathcal{D}_{AB}Sc^{1/3}(\rho V)^{1/2}}{\mu^{1/2}L}(2L^{1/2}) = \frac{(0.2155)(35100)cm^2(1.7153)^{1/3}(7.308E\text{-}6)^{1/2}kg^{1/2}(100)^{1/2}cm^{1/2}(2)(10)^{1/2}cm^{1/2}m^{1/2}s}{s\ m^{3/2}s^{1/2}(44E\text{-}5)^{1/2}s^{1/2}kg^{1/2}(10)cm}$

$= 23340\ \frac{cm^2}{ms}\frac{(1)m}{(100)cm} = \boxed{233\ \frac{cm}{s} = \bar{k}_c}$

**g)** Using $\bar{k}_c$ from (e), calculate the total rate of diamond film formation (mol/min).

g) $N_{Ay} = \bar{k}_c(c_{Am} - c_{As}) = \bar{k}_c c_{A\infty} \rightarrow n_{CH_4} = n_{form} = \bar{k}_c c_{A\infty} \times L^2 = (233)\frac{cm}{s}(0.005)\left(3.6559E\text{-}3\right)\frac{mol}{m^3}(10^2)cm^2 \times \frac{(1)^3 m^3}{(100)^3 cm^3} \times \frac{(60)s}{(1)min}$

($N_{Ay}$: flux of $CH_4$ to wafer; $c_{Am}$: in bulk; $c_{As} \approx 0$ (rapid reaction); $n_{CH_4}$: moles $CH_4$ consumed/time; $c_{tot}$ (i.g))

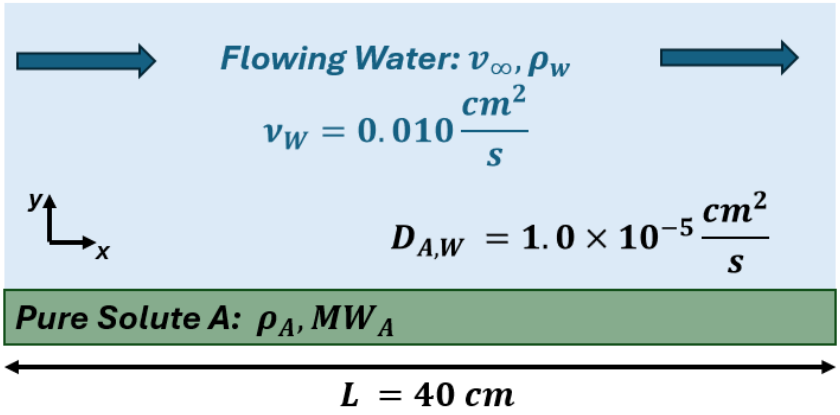
$\boxed{n_{form} = 2.54E\text{-}5\ \frac{mol}{min}}$

**h)** For a boundary layer developed by a laminar flow, the local Sherwood number can be calculated as $0.323Re_x^{1/2}Sc^{1/3}$. Why does the prefactor magnitude decrease relative to the film theory estimate?

h) Film theory assumes that the diffusion layer is stagnant (it is not completely stationary).

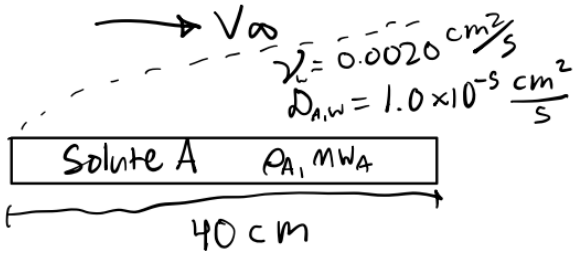
## Problem 4 (XX pts)

Liquid water is flowing over a flat square slab of pure solute A. As the liquid flows over the slab, which has length $L = 40\ cm$ and width $W = 40\ cm$, solute A dissolves into the liquid. For this process, your goal is to maintain laminar flow over the slab. Solute A has a solubility of 2 $\frac{mol}{L}$ in water. It is helpful to know that the diffusivity of A through water is $D_{A,W} = 1.0 \times 10^{-5} \frac{cm^2}{s}$ and the kinematic viscosity of water is $\nu_W = 0.010 \frac{cm^2}{s}$.




**a)** What is the highest bulk velocity that will maintain a laminar boundary layer?

Problem 4 – Solute A dissolving into water




$.000001 \frac{m^2}{s} \times \left(\frac{100}{1} \frac{cm}{m}\right)^2$

a. Highest bulk velocity which maintains Laminar Flow

Transition laminar => turbulent

certain for $Re < 2 \times 10^5$

$$Re = \frac{L V_\infty}{\nu_w}$$

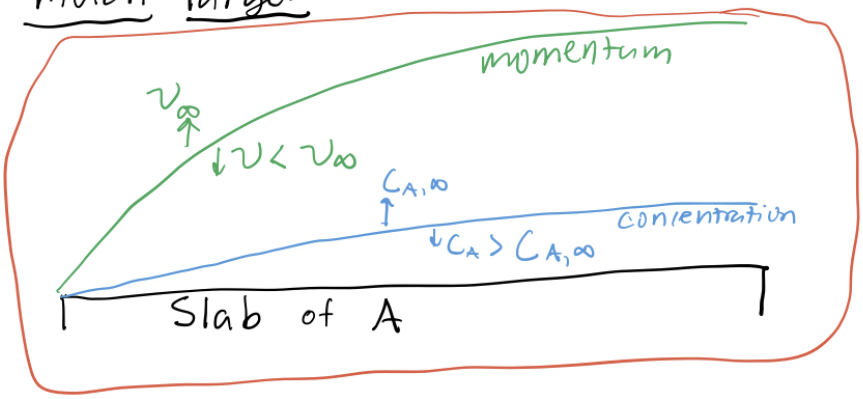
$$V_{\infty,crit} = \frac{Re_{crit} \nu_w}{L} = \frac{2 \times 10^5 \cdot 0.010 \frac{cm^2}{s}}{40\ cm} = 50 \frac{cm}{s}$$

**b)** Sketch the relative concentration and hydrodynamic boundary layers. Which nondimensional term compares their magnitude?

b. hydrodynamic vs concentration boundary layer.

Schmidt #: $Sc = \frac{\text{momentum diffusivity}}{\text{mass diffusivity}} = \frac{\nu_w}{D_{A,w}} = \frac{0.010 \frac{cm^2}{s}}{0.000010 \frac{cm^2}{s}}$

$Sc = 1000$, so momentum boundary layer is much larger than concentration number



**c)** Calculate the film-transfer coefficient $k_c$ at a distance of 10 $cm$ and 30 $cm$ from the leading edge of the slab. At which point is it larger and why?

C Calculate film transfer coefficient @ 10 + 30 cm

Schmidt #: $Sc = \frac{\nu_w}{D_{AW}} = \frac{0.010 \frac{cm^2}{s}}{0.000010 \frac{cm^2}{s}} = 1000$

Reynolds #:

10 cm: $Re_{x,10} = \frac{x v_\infty}{\nu} = \frac{(10\ cm)(50 \frac{cm}{s})}{(.010 \frac{cm^2}{s})} = 50{,}000$

30 cm $Re_{x,30} = \frac{x v_\infty}{\nu} = \frac{(30\ cm)(50 \frac{cm}{s})}{(.010 \frac{cm^2}{s})} = 150{,}000$

Sherwood #:

$Sh_{x,10} = 0.332\, Re_{x,10}^{1/2} Sc^{1/3} = 0.332\,(50{,}000)^{1/2}(1000)^{1/3} = 742.4$

$Sh_{x,30} = 0.332\, Re_{x,30}^{1/2} Sc^{1/3} = 0.332\,(150{,}000)^{1/2}(1000)^{1/3} = 1286$

get mass transfer coefficients: $Sh_x = \frac{k_{c,x} X}{D_{AB}}$

$k_{c,x} = \frac{Sh_L D_{AB}}{X}$

$k_{c,10} = \frac{Sh_{10} D_{A,W}}{10 cm} = \frac{(742.4)(0.000010 \frac{cm^2}{s})}{10\ cm} = 7.42 \times 10^{-4} \frac{cm}{s}$

$k_{c,30} = \frac{Sh_{30} D_{A,W}}{30 cm} = \frac{(1286)(0.000010 \frac{cm^2}{s})}{30\ cm} = 4.29 \times 10^{-4} \frac{cm}{s}$

$k_c$ is higher at 10 cm b/c there is a thinner boundary layer + thus a higher local gradient in concentration to drive transfer.

**d)** Calculate the average mass transfer coefficient for the slab.

d. Average Mass transfer coefficient

$$\bar{k}_c = \frac{\int_0^L k_c\,dx}{\int_0^L dx} = \frac{\int_0^L \frac{\mathcal{D}_{A,w}}{x} Sh_x}{L} = \frac{\int_0^L 0.332 \frac{\mathcal{D}_{A,w}}{x} Re^{1/2} Sc^{1/3}\,dx}{L} = \frac{\int_0^L \frac{0.332\mathcal{D}_{A,w}}{x}\left(\frac{v_\infty x}{\nu}\right)^{1/2} Sc^{1/3}\,dx}{L}$$

$$= \frac{0.332\mathcal{D}_{A,w} Sc^{1/3}\left(\frac{v_\infty}{\nu}\right)^{1/2}}{L} \int_0^L x^{-1/2}\,dx = \frac{0.332\mathcal{D}_{A,w} Sc^{1/3}\left(\frac{v_\infty}{\nu}\right)^{1/2}}{L} \cdot 2(L^{1/2})$$

$$= \frac{0.664\mathcal{D}_{A,w}}{L} Re_L^{1/2} Sc^{1/3}$$

$$= 0.664 \cdot \frac{1\times10^{-5}\ \frac{cm^2}{s}}{40\ cm}\left(\frac{50\ \frac{cm}{s}\cdot 40\ cm}{0.010\ \frac{cm^2}{s}}\right)^{1/2}(1000)^{1/3}$$

$$\bar{k}_c = 7.42\times10^{-4}\ \frac{cm}{s}$$

**e)** What is net molar flux of A dissolving into the flowing water?

e. Net flux of A dissolving into the flowing water

$$N_A = k_{c,avg}(C_{A,s} - C_{A,\infty}) = k_{c,avg} C_{A,s}$$

$$N_A = \left(7.42\times10^{-4}\frac{cm}{s}\right)\left(2.0\ \frac{mol}{L} \times \frac{1\ L}{1000\ cm^3}\right) = 1.48\times10^{-6}\ \frac{mol}{cm^2\cdot s}$$

$$\text{net flux of A} = N_A \cdot A_{slab} = 1.48\times10^{-6}\ \frac{mol}{cm^2\cdot s}\left(40\ cm \times 40\ cm\right)$$

$$\text{net flux of A} = 2.38\times10^{-3}\ \frac{mol}{s}$$

**f)** Where is the maximum local flux of A dissolving into water? Why?

f) Local Flux of A is maximized at the leading edge of the slab. This is because the local mass transport coefficient is highest. This is a result of boundary layer theory, because the boundary is infinitessimal & thus the driving force for mass transport approaches infinity